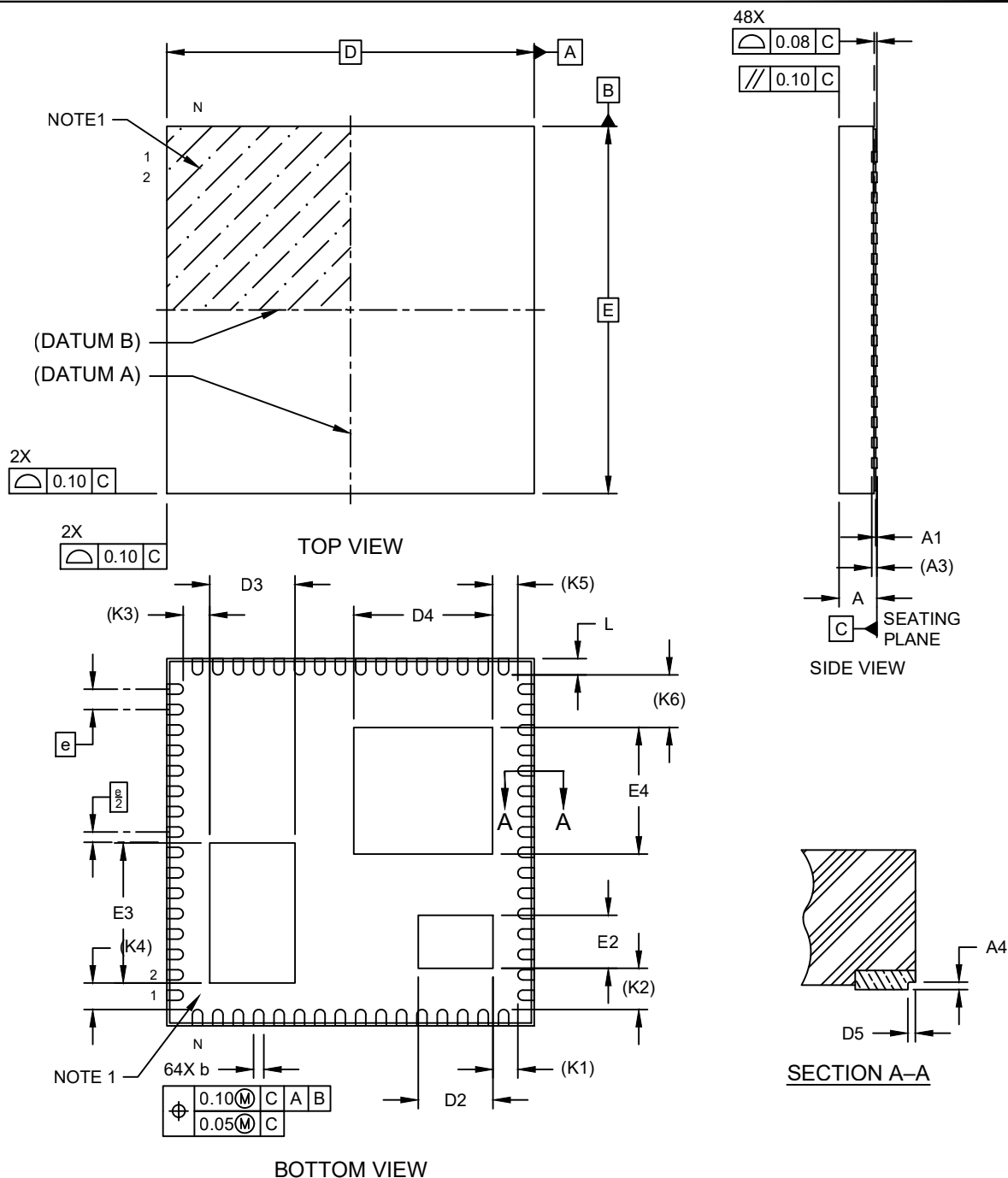


Packaging Diagrams and Parameters

64-Lead Very Thin Grid Array Quad Flat, No Lead Package (2JW) - 9x9x0.927 mm Body [VGQFN]
With Multiple Exposed Pads and Stepped Wettable Flanks

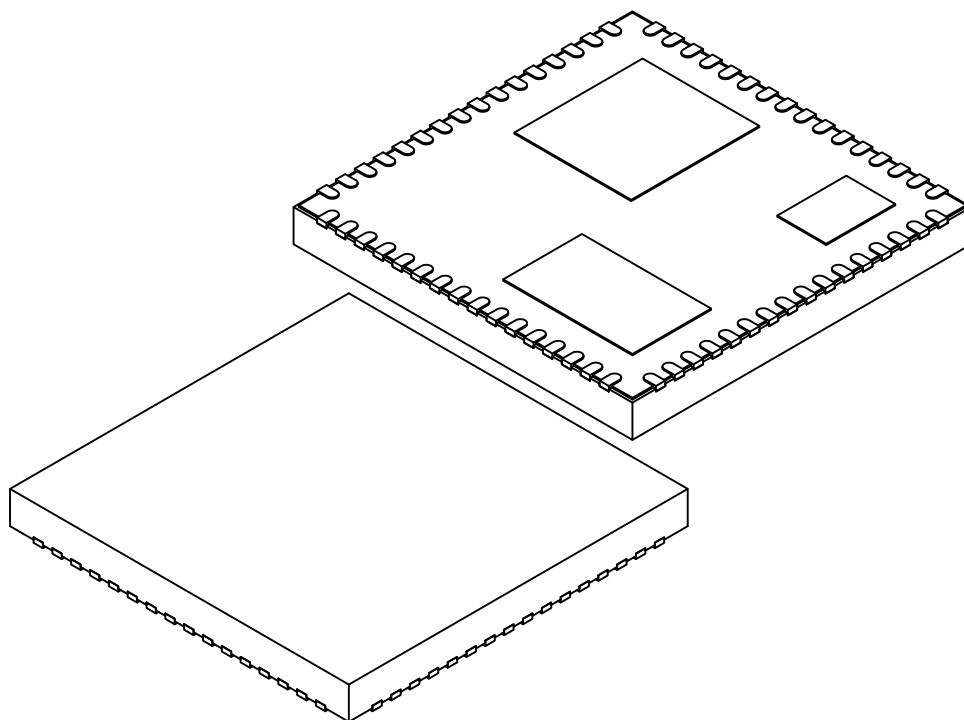
Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Packaging Diagrams and Parameters

64-Lead Very Thin Grid Array Quad Flat, No Lead Package (2JW) - 9x9x0.927 mm Body [VGQFN]
With Multiple Exposed Pads and Stepped Wettable Flanks

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	64		
Pitch	e	0.50 BSC		
Overall Height	A	0.827	0.877	0.927
Standoff	A1	0.00	-	0.05
Terminal Thickness	A3	0.127 REF		
Overall Length	D	9.00 BSC		
Exposed Pad Length	D2	1.73	1.83	1.93
Exposed Pad Length	D3	1.99	2.09	2.19
Exposed Pad Length	D4	3.30	3.40	3.50
Overall Width	E	9.00 BSC		
Exposed Pad Width	E2	1.20	1.30	1.40
Exposed Pad Width	E3	3.33	3.43	3.53

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Exposed Pad Width	E4	3.00	3.10	3.20
Terminal Width	b	0.20	0.25	0.30
Terminal Length	L	0.30	0.40	0.50
Terminal-to-Exposed-Pad	K1	0.61 REF		
Terminal-to-Exposed-Pad	K2	1.01 REF		
Terminal-to-Exposed-Pad	K3	0.65 REF		
Terminal-to-Exposed-Pad	K4	0.65 REF		
Terminal-to-Exposed-Pad	K5	0.62 REF		
Terminal-to-Exposed-Pad	K6	1.29 REF		
Wettable Flank Step Cut Length	D5	0.01	0.05	0.09
Wettable Flank Step Cut Height	A4	0.03	-	-

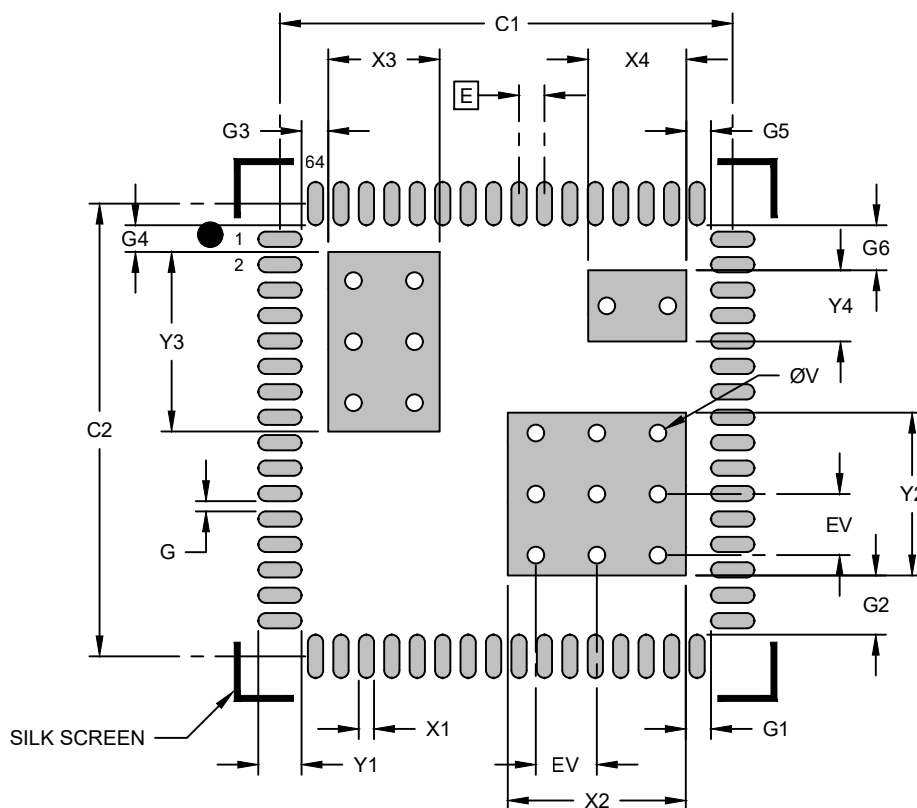
Notes:

1. The Pin 1 visual index feature may vary, but it must be located within the hatched area.
2. The package is saw singulated.
3. Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

Land Pattern

64-Lead Very Thin Grid Array Quad Flat, No Lead Package (2JW) - 9x9x0.927 mm Body [VGQFN]
With Multiple Exposed Pads and Stepped Wettable Flanks

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS			Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX	Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.50 BSC			Contact Pad Length	Y1			0.85
Contact Pad Spacing	C1		8.90		Contact Pad to Contact Pad	G	0.20		
Contact Pad Spacing	C2		8.90		Contact Pad to Center Pad	G1	0.50		
Center Pad Width	X2			3.50	Contact Pad to Center Pad	G2	1.17		
Center Pad Length	Y2			3.20	Contact Pad to Center Pad	G3	0.53		
Center Pad Width	X3			2.19	Contact Pad to Center Pad	G4	0.53		
Center Pad Length	Y3			3.53	Contact Pad to Center Pad	G5	0.49		
Center Pad Width	X4			1.93	Contact Pad to Center Pad	G6	0.89		
Center Pad Length	Y4			1.40	Thermal Via Diameter	V		0.33	
Contact Pad Width	X1			0.30	Thermal Via Pitch	EV		1.20	

Notes:

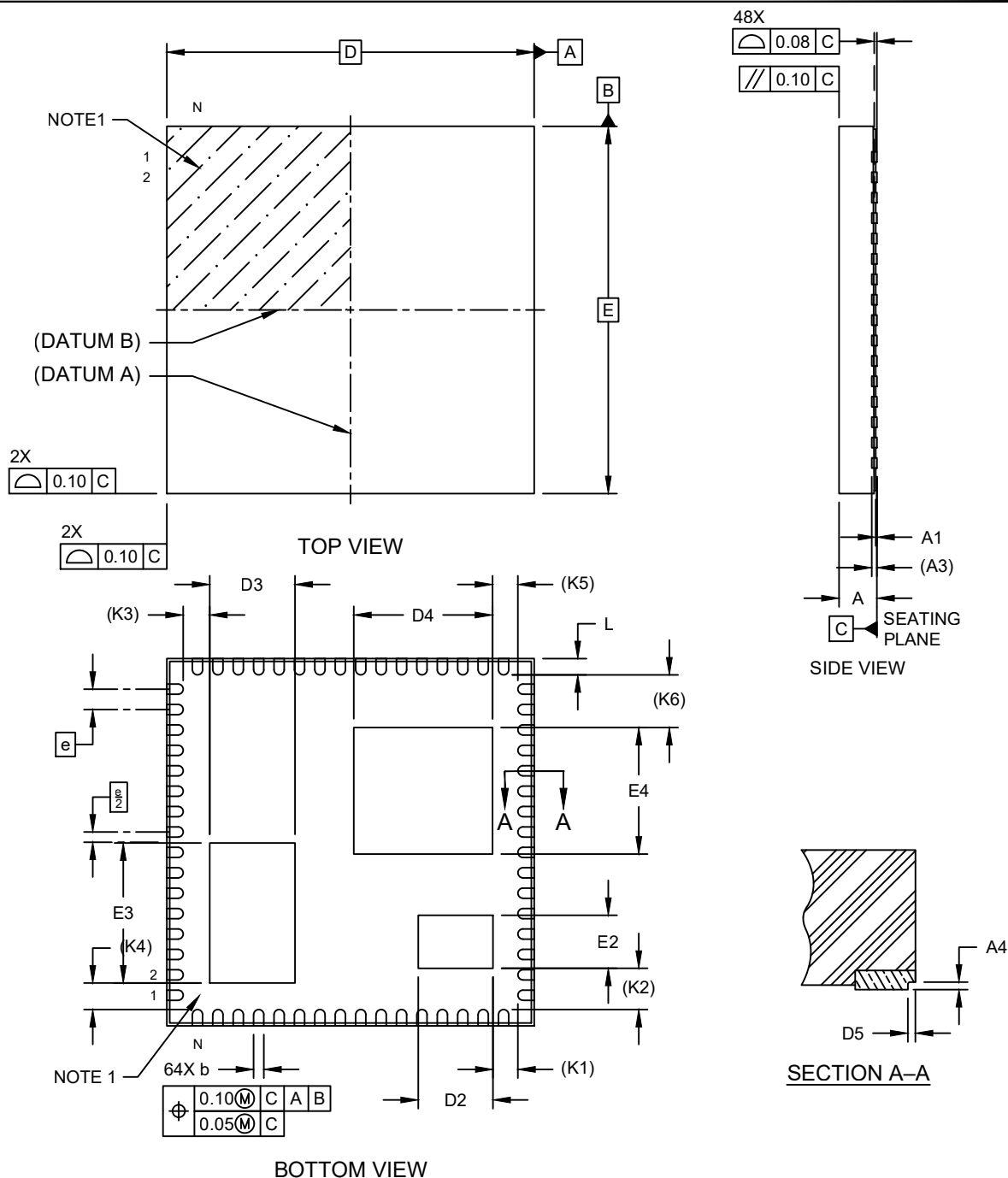
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during the reflow process.

Microchip Technology Drawing C04-02544-2JW Rev E

Packaging Diagrams and Parameters

64-Lead Very Thin Grid Array Quad Flat, No Lead Package (M9) - 9x9x0.927 mm Body [VGQFN]
With Multiple Exposed Pads and Stepped Wettable Flanks

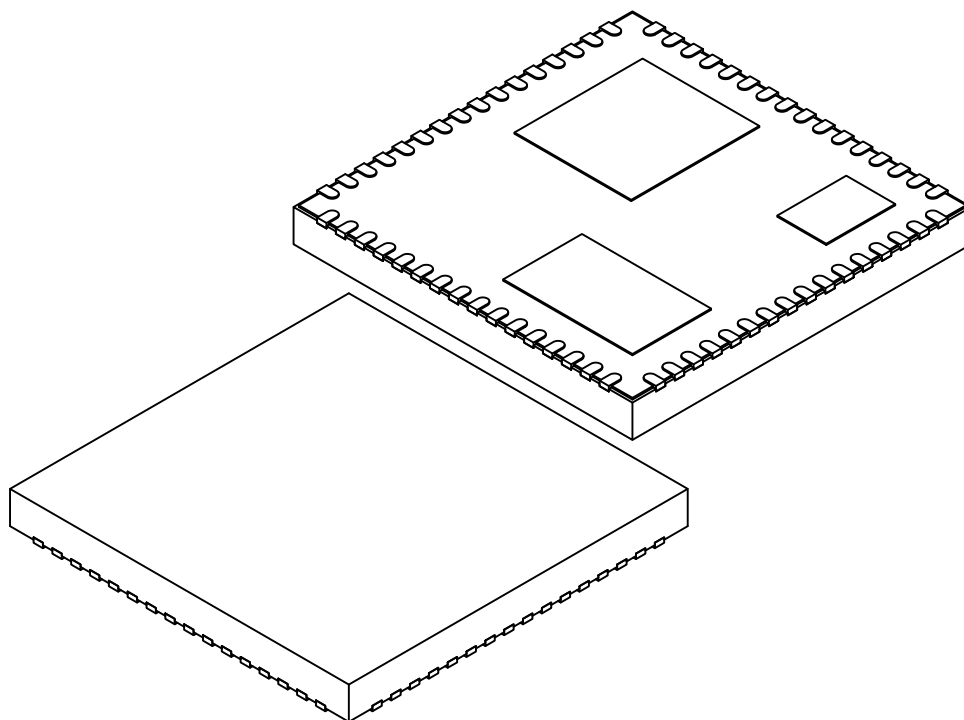
Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Packaging Diagrams and Parameters

64-Lead Very Thin Grid Array Quad Flat, No Lead Package (M9) - 9x9x0.927 mm Body [VGQFN]
With Multiple Exposed Pads and Stepped Wettable Flanks

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	64		
Pitch	e	0.50 BSC		
Overall Height	A	0.827	0.877	0.927
Standoff	A1	0.00	-	0.05
Terminal Thickness	A3	0.127 REF		
Overall Length	D	9.00 BSC		
Exposed Pad Length	D2	1.73	1.83	1.93
Exposed Pad Length	D3	1.99	2.09	2.19
Exposed Pad Length	D4	3.30	3.40	3.50
Overall Width	E	9.00 BSC		
Exposed Pad Width	E2	1.20	1.30	1.40
Exposed Pad Width	E3	3.33	3.43	3.53

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Exposed Pad Width	E4	3.00	3.10	3.20
Terminal Width	b	0.20	0.25	0.30
Terminal Length	L	0.30	0.40	0.50
Terminal-to-Exposed-Pad	K1	0.61 REF		
Terminal-to-Exposed-Pad	K2	1.01 REF		
Terminal-to-Exposed-Pad	K3	0.65 REF		
Terminal-to-Exposed-Pad	K4	0.65 REF		
Terminal-to-Exposed-Pad	K5	0.62 REF		
Terminal-to-Exposed-Pad	K6	1.29 REF		
Wettable Flank Step Cut Length	D5	0.01	0.05	0.09
Wettable Flank Step Cut Height	A4	0.03	-	-

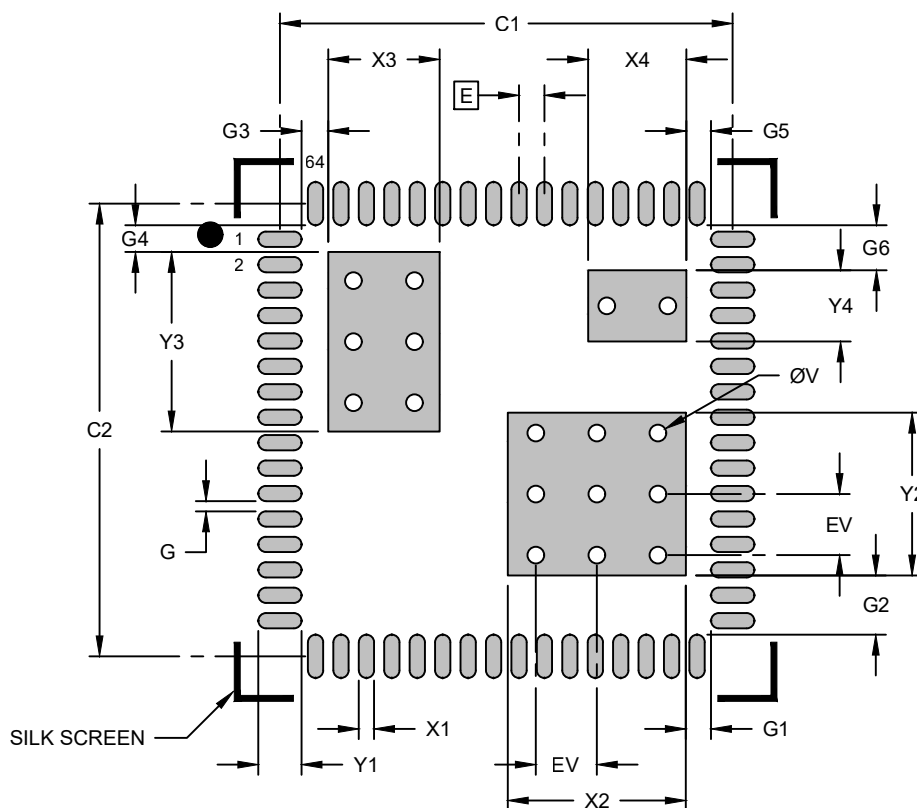
Notes:

1. The Pin 1 visual index feature may vary, but it must be located within the hatched area.
2. The package is saw singulated.
3. Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

Land Pattern

64-Lead Very Thin Grid Array Quad Flat, No Lead Package (M9) - 9x9x0.927 mm Body [VGQFN]
With Multiple Exposed Pads and Stepped Wettable Flanks

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS			Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX	Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.50 BSC			Contact Pad Length	Y1			0.85
Contact Pad Spacing	C1		8.90		Contact Pad to Contact Pad	G	0.20		
Contact Pad Spacing	C2		8.90		Contact Pad to Center Pad	G1	0.50		
Center Pad Width	X2			3.50	Contact Pad to Center Pad	G2	1.17		
Center Pad Length	Y2			3.20	Contact Pad to Center Pad	G3	0.53		
Center Pad Width	X3			2.19	Contact Pad to Center Pad	G4	0.53		
Center Pad Length	Y3			3.53	Contact Pad to Center Pad	G5	0.49		
Center Pad Width	X4			1.93	Contact Pad to Center Pad	G6	0.89		
Center Pad Length	Y4			1.40	Thermal Via Diameter	V		0.33	
Contact Pad Width	X1			0.30	Thermal Via Pitch	EV		1.20	

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during the reflow process.

Microchip Technology Drawing C04-02544-M9 Rev E